

LLM-WER / LLM-CER

A better error metric for Indian-language ASR

The problem with standard WER/CER

Word Error Rate (WER) and Character Error Rate (CER) were built around English: one accepted spelling per word, minimal mid-sentence language mixing, and a small gap between formal and spoken usage. Indian languages routinely break all three assumptions, so a plain string-diff between reference and prediction ends up penalizing transcriptions that are, to a native speaker, correct.

Sarvam's analysis identifies six recurring failure classes:

#	Failure class	Example	What plain WER/CER does
1	Colloquial variants	Tamil ASR outputs “avunga” instead of the formal “avargal”	Counts a full error, even though a native speaker hears a perfect transcription
2	Code-mixing / script mismatch	डॉक्टर vs. doctor	Flags it as a substitution, even though both spellings are correct
3	Short helper words	नहीं नहीं है instead of नहीं है	One stray helper word inflates the score out of proportion to the mistake
4	Agglutinative morphology	A small suffix change on a long Malayalam/Telugu compound	Creates a large CER penalty for a negligible difference
5	Numeric format variability	पांच सौ / 500 / ५००	Treats all three as unrelated tokens
6	Severity blindness	Inserting नहीं ("not") reverses the whole meaning	Scores it the same low-weight way as any other one-word substitution — it can't tell a meaning-flipping error from a trivial one

Note that class 6 cuts the other way: WER isn't too harsh here, it's too flat — it can't distinguish a meaning-reversing insertion from a harmless one, so both get the same small penalty.

The fix: LLM-WER and LLM-CER

LLM-WER/CER keep the standard alignment step, but hand every mismatched segment to an LLM before scoring it as an error:

- Run standard WER/CER alignment to locate every mismatched segment between reference and prediction.
- Ask an LLM, per segment: is this the same meaning? Is it phonetically/semantically the same word?
- Only segments the LLM confirms are genuinely wrong count against the score; everything else is treated as correct.

Validation on a 12-example Hindi set

Running the pipeline on a small hand-built test set covering all six failure classes plus two genuine ASR errors:

Reference	Prediction	Class	Orig. WER	LLM-WER
वह धीरे-धीरे चल रहा है	वह धीरे धीरे चल रहा है	Hyphenation	33.3%	0%
उन्नीस सौ नब्बे में यह हुआ	1990 में यह हुआ	Numeric (5)	50.0%	0%
मुझे सौ रुपये चाहिए	मुझे ₹100 चाहिए	Currency (5)	50.0%	0%
मैंने पी एन बी में खाता खोला	मैंने पीएनबी में खाता खोला	Acronym	42.9%	0%
मैंने अमेज़न से ऑर्डर किया	मैंने Amazon से ऑर्डर किया	Script (2)	20.0%	0%
यह मेज़ पर रखा है	यह मेज़ पे रखा है	Contraction	20.0%	0%
मैं क्या कर रहा हूँ	मैं का कर रिया हूँ	Real ASR error	40.0%	40.0%
वह डॉक्टर के पास गया	वह doctor के पास गया	Script (2)	20.0%	0%
यह सही नहीं है	यह सही नहीं नहीं है	Helper word (3)	20.0%	20.0%
भारत की स्वतंत्रता का दिन	भारत की स्वतन्त्रता का दिन	Spelling variant	20.0%	0%
पांच सौ रुपये	५०० रुपये	Numeric (5)	66.7%	0%
मुझे यह पसंद है	मुझे यह नहीं पसंद है	Severity (6)	20.0%	20.0%

Mean original WER: 33.6% → Mean LLM-WER: 6.7%

Key insight: plain WER/CER often penalizes the ASR model for no real error at all. वह धीरे-धीरे चल रहा है vs. वह धीरे धीरे चल रहा है is the same word with different spacing, scored as a mistake anyway. Standard metrics can't tell "the model got it right, just wrote it differently" apart from "the model actually got it wrong" — every one of the six failure classes above is really the same root problem wearing a different disguise. That's the gap LLM-WER/CER exists to close.

References

Blog- <https://www.sarvam.ai/blogs/evaluating-indian-language-asr>

Repo- https://github.com/sarvamai/llm_wer